Project Report

Student Details

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About Me: I am a Diploma-level student at IIT Madras, pursuing BS in Data Science and Applications. I have a strong passion for application development, backend engineering, graphic design and UI/UX design. I love creating real-world solutions that blend clean interfaces with functional, reliable backend systems.

Project Details

Project title : TrekMate - Trekking Management Application V2
Problem statement : You are required to build a Trekking Management Application (TMA) web application that allows Admin, Trek Staff, and Users (Trekkers) to interact with the system based on their roles.
Approach: The project follows a client-server architecture using Vue.js for the frontend and Flask REST APIs for the backend. Authentication is implemented using JWT Tokens and Werkzeug, while SQLAlchemy manages database operations. Celery and Redis are used for background tasks such as automated monthly reports and CSV export generation. Bootstrap provides a responsive and user-friendly interface throughout the application.

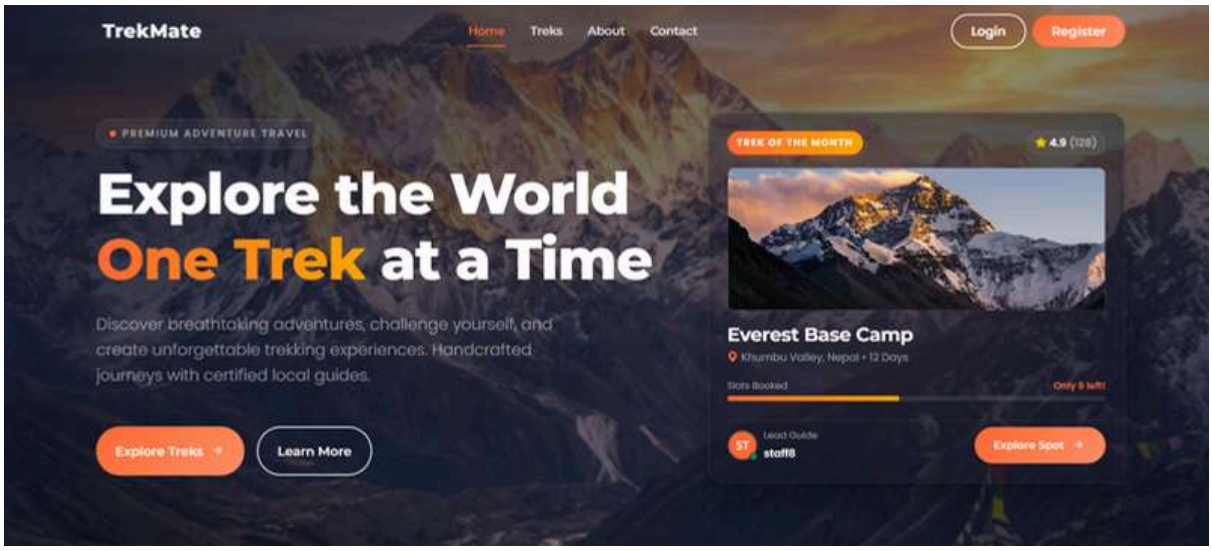
AI / LLM Declaration

I, Ridhi Sehgal, declare that this project was primarily developed through my own learning, coding, implementation, and debugging efforts.

AI tools such as ChatGPT, Copilot, Claude were used only for:

- Debugging Python and Vue.js errors
- Understanding Flask, JWT, SQLAlchemy, and Celery concepts
- Improving documentation and report writing
- UI/UX improvements
- Solving specific implementation issues

Approximately **15–20%** of the project involved AI-assisted learning and debugging. The overall system design, database schema, REST API implementation, frontend integration, authentication, and application logic were independently developed by me.



Landing Page of TrekMate

The homepage provides an overview of the trekking management system, showcasing featured treks, key services, navigation menu, and essential information for users to explore and access the platform.

Technologies and Frameworks

Technology	Purpose
Flask	REST API Backend
Vue.js 3	Frontend SPA
Vite	Frontend Development Server
SQLAlchemy	ORM for Database
SQLite	Relational Database
Flask JWT Extended	Authentication using JWT
Flask CORS	API Communication
Bootstrap 5	Responsive UI
Axios	API Requests
Celery	Background Task Processing
Redis	Celery Broker and Result Backend
SMTP / Mailhog	Automated Email Reports

YAML API Definition File: Included separately in the submission ZIP as **api.yaml**.

Database Schema / ER Diagram

Tables

1. User

- Stores user information, authentication details, role, and profile.

2. Trek

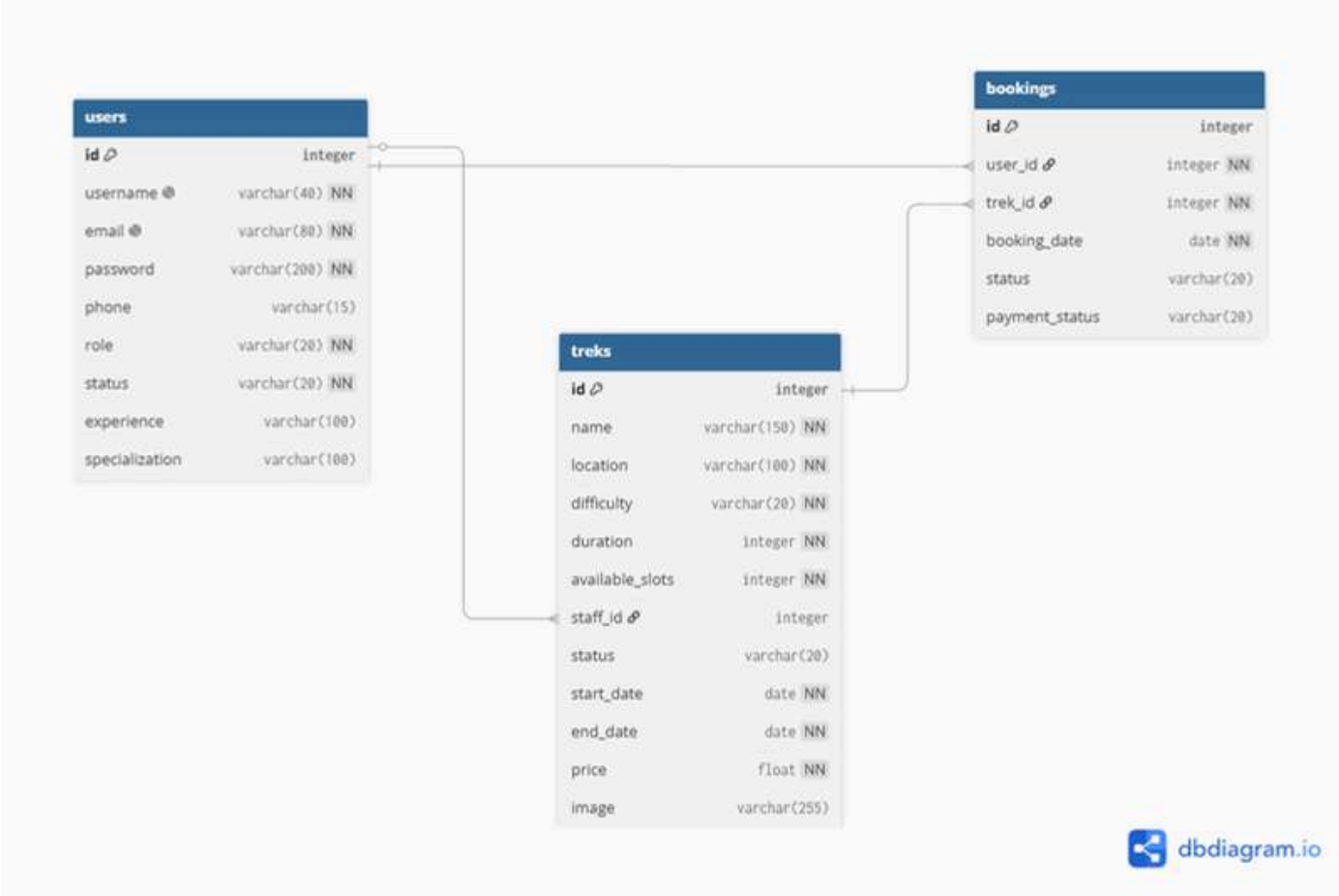
- Stores trekking package information including location, duration, difficulty, slots, guide assignment, and pricing.

3. Booking

- Stores booking records linking users and treks along with booking and payment status.

Relationships

- One-to-Many → User → Trek (One staff member can manage multiple treks.)
- One-to-Many → User → Booking (One user can make multiple bookings.)
- One-to-Many → Trek → Booking (One trek can have multiple bookings.)



API Resource Endpoints

Endpoint	Method	Description
/register	POST	Register a new user
/login	POST	Authenticate user and generate JWT token
/profile	GET / PUT	View and update user profile
/treks	GET	Retrieve available trekking packages

/book	POST	Book a trekking package
/booking/history	GET	View user's booking history
/admin/users	GET	Manage registered users
/admin/treks	GET	Manage trekking events
/admin/bookings	GET	View all bookings

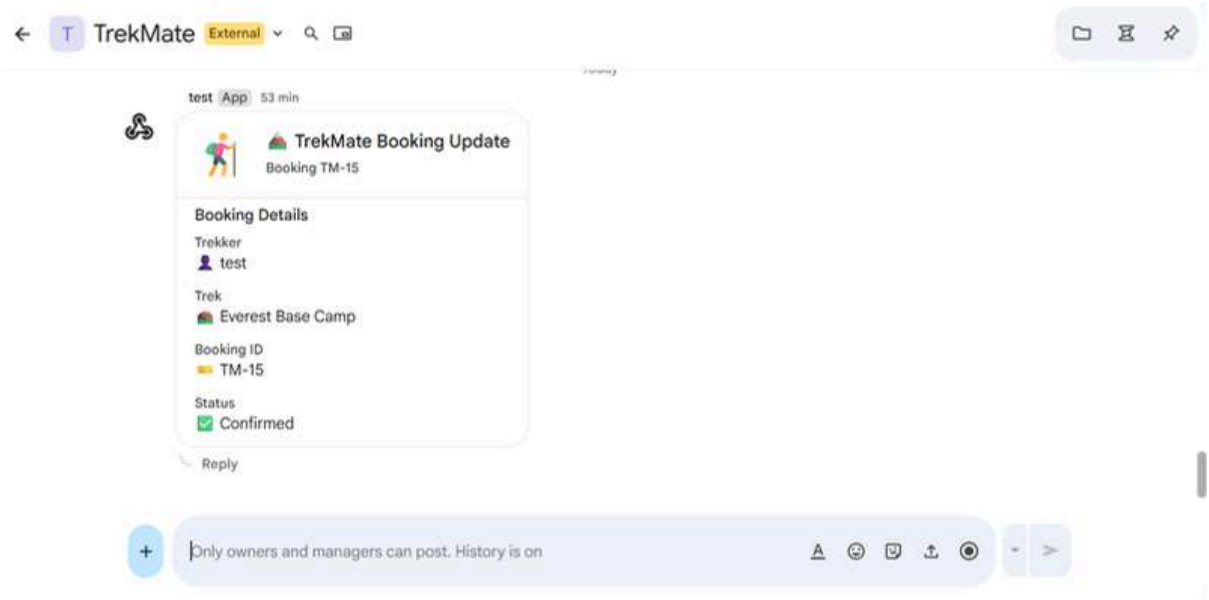
Architecture and Features

Architecture Overview

- Vue.js – Frontend user interface
- Axios – API communication
- Flask – REST API backend
- SQLAlchemy – Database models and queries
- SQLite – Data storage
- JWT – Secure authentication
- Celery + Redis – Background task scheduling
- MailHog – Email testing
- Google Chat Webhooks – Real-time notifications

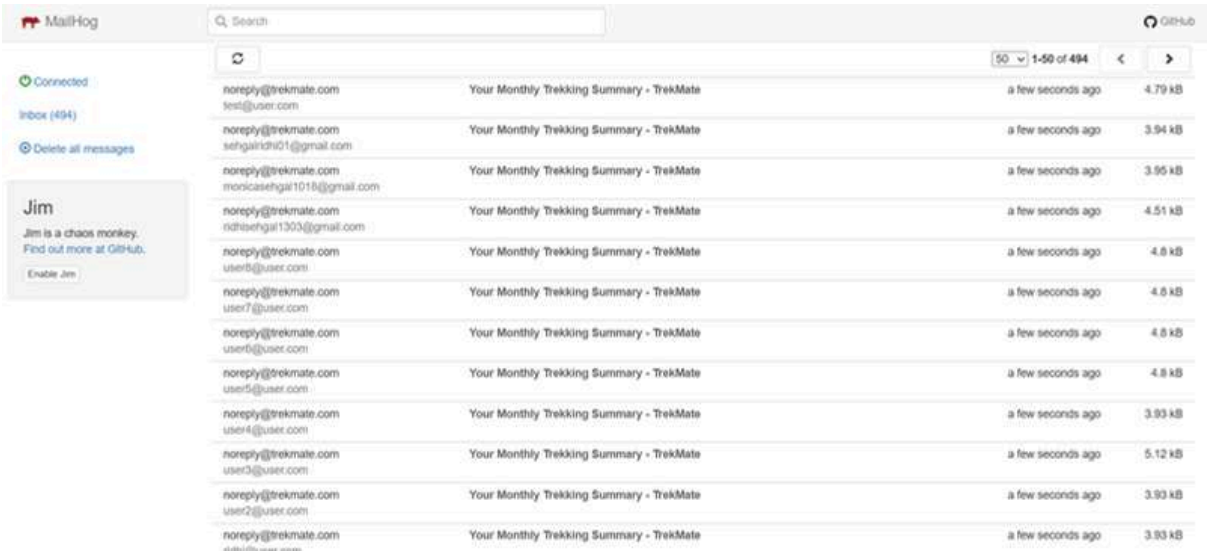
Implemented Features

- User registration and secure JWT login
- Role-based access for Admin, Staff, and Users
- Trek management and guide assignment
- Trek booking and booking history
- Profile management
- Automated monthly email reports using Celery Beat
- CSV export of booking reports
- Google Chat webhook notifications for booking updates
- Responsive user interface using Bootstrap and Vue.js



Webhooks Implementation using GChat

Successful delivery of a real-time booking update to the connected Google Chat



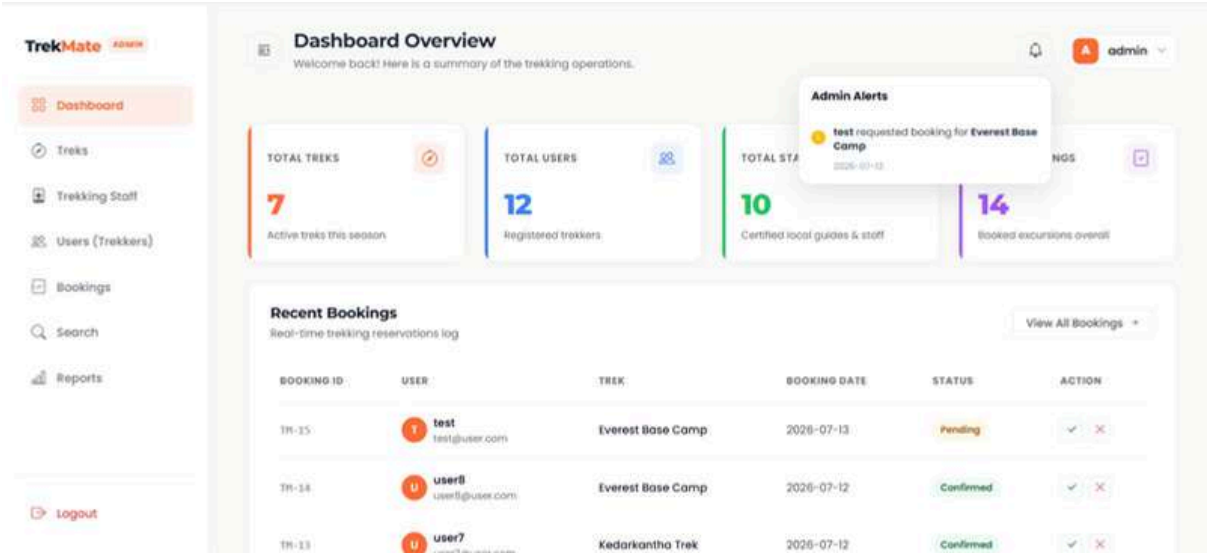
Automated Email reports on MailHog

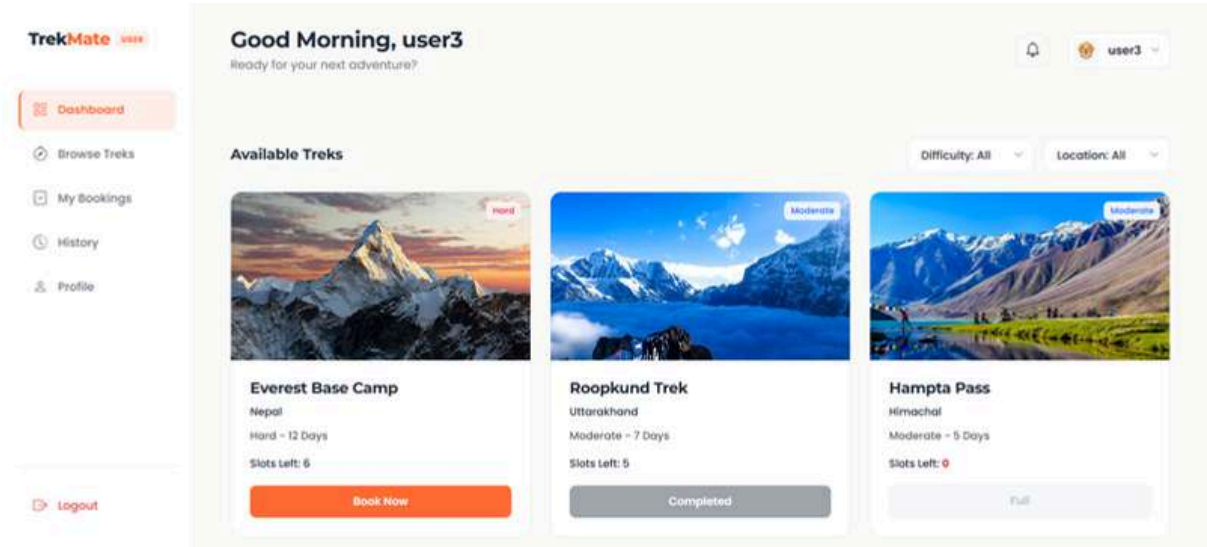
MailHog displays monthly trekking summary emails automatically generated and sent to registered users by Celery Beat's scheduled tasks.

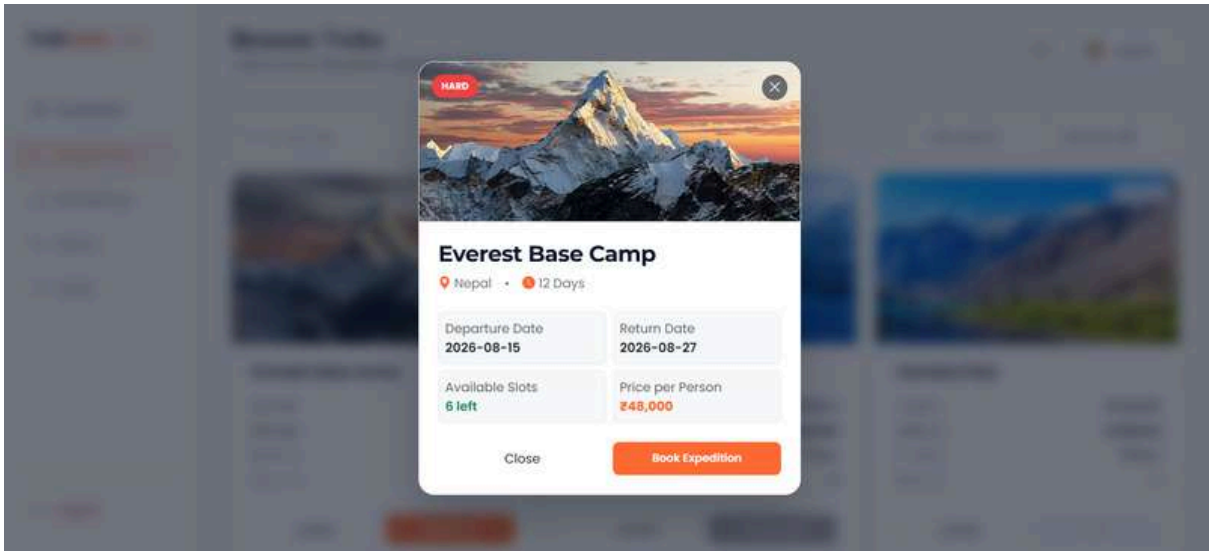
Video Presentation

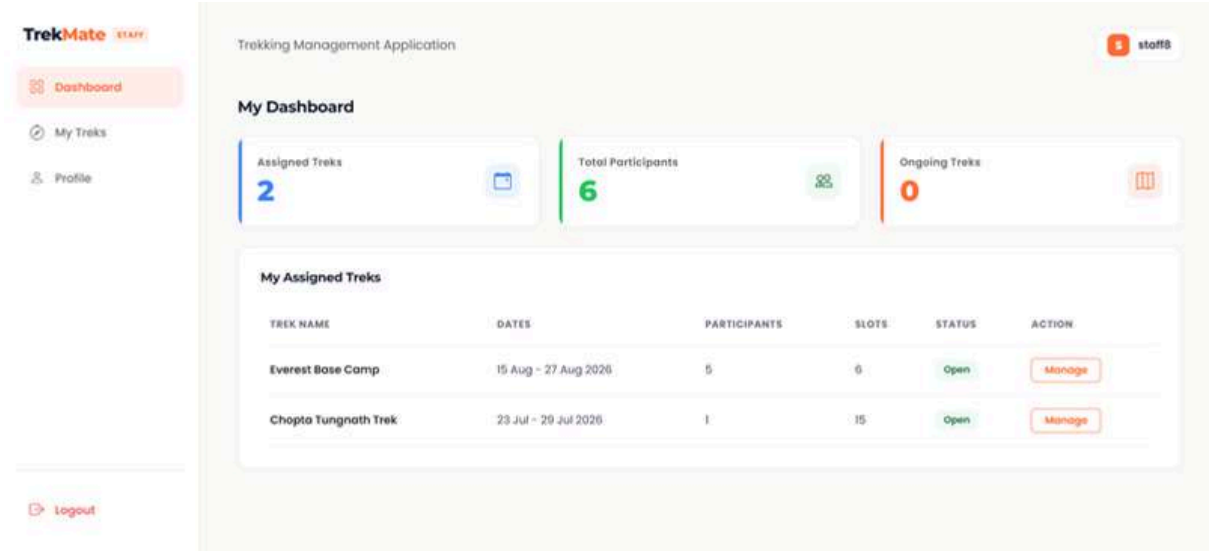
https://drive.google.com/drive/folders/17RU9IMUsPhEtHHud9s6vVxB64WtxEqyV?usp=sharing

(Accessible to all with “View” Access)






HARD
Everest Base Camp
Nepal • 12 Days
Departure Date
2026-08-15
Return Date
2026-08-27
Available Slots
6 left
Price per Person
₹48,000
Close
Book Expedition


TrekMate STAFF
Dashboard
My Treks
Profile
Logout
Trekking Management Application
staff8
My Dashboard
Assigned Treks
2
Total Participants
6
Ongoing Treks
0
My Assigned Treks
TREK NAME
DATES
PARTICIPANTS
SLOTS
STATUS
ACTION
Everest Base Camp
15 Aug - 27 Aug 2026
5
6
Open
Manage
Chopta Tungnath Trek
23 Jul - 29 Jul 2026
1
15
Open
Manage


Thank you